

RESEARCH ARTICLE

Dopamine-Assisted Rapid Construction of Zwitterionic Polymeric Coating for Improved Antifouling and Bactericidal Properties

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ABSTRACT

With the escalating challenges of biological contamination, surface modification technology has become a promising strategy to develop multifunctional materials. Antifouling coatings based on zwitterionic/quaternary ammonium polymers have been widely applied in medical, chemical, and marine antifouling fields due to their excellent biological inertness and remarkable antifouling and bactericidal properties. However, the limited adhesion of these coatings to substrate surfaces significantly restricts their application. An ideal antifouling and bactericidal coating should not only exhibit high antifouling and bactericidal efficiency but also form robust interfacial bonds with the substrate. Mussel-inspired dopamine (DA) functional polymers with superior adhesion and multi-functionality have been extensively studied and applied in fields such as bio-adhesives, surface modification, and biomedicine. One-step co-deposition of PDA with antifouling polymers, especially zwitterionic polymers, has been considered as a promising strategy for fabricating antifouling surfaces. However, the key challenge lies in the slow co-deposition process, which typically requires 10 h or even longer. In this work, a rapid, stable, and universal method utilizing CuSO_4 and H_2O_2 to catalyze the co-deposition of DA and zwitterionic polymer was proposed, which enables the formation of a stable coating within 45 min. The copolymer of glycidyl methacrylate (GMA), [2-(methacryloyloxy) ethyl] dimethyl-(3-sulfopropyl) ammonium hydroxide (DMAPS), and [2-(methacryloyloxy) ethyl] trimethylammonium chloride solution (DMC) (P(GMA-co-DMAPS-co-DMC)) developed in this study achieves synergistic antifouling and bactericidal effects through a simple and efficient grafting method, demonstrating significant potential for biomedical and industrial applications. The results demonstrate that the grafted coating exhibits excellent anti-protein adsorption performance (over 90%) as well as significant bactericidal performance, with the inactivation rates of 99.9% and 97.7% against *Staphylococcus aureus* and *Escherichia coli*, respectively. Additionally, the coating exhibits strong adhesion to the substrate, further validating its practical application. New insights and methods will be provided for the development of high-performance antifouling and bactericidal coatings.

1 | Introduction

Surface pollution and bacterial adhesion are critical issues that seriously affect the performance and lifespan of materials in the fields of marine engineering, biomedical materials, and water treatment [1–3]. In marine environments, biological pollution has a negative impact on ships, marine transportation pipelines, and some marine engineering structures [4]. Marine biological adhering to the hull not only increases its weight and roughness but also causes the vessel to slow down and lose maneuverability, thereby escalating fuel consumption and the emission of harmful compounds [5]. For instance, nuclear power plants commonly use seawater to dissipate heat. The fouled organisms are prone to adhere to the inner wall of their pipelines, which reduces the inner diameter of the pipeline and increases the roughness, which in turn decreases the water inlet and affects the heat dissipation of the system [6]. In addition, some defiled organisms may fall off and enter the rotating filter, affecting work efficiency, or even stopping working. In the field of biomedical science, biological pollution can damage medical devices, dressings, etc. In sensors, protein adsorption will reduce sensor sensitivity and distort the data, further affecting accuracy [7]. The adhesion of proteins and platelets on various catheters and hemodialysis equipment often affects the safety of patients. Up to date, a great deal of work has focused on the increasingly severe issue of biofouling on material surfaces. The preparation of coatings on material surface is a typical instance. The most widely studied coatings currently include antifouling coatings and sterilizing coatings [8], depending on the effect of these surfaces on the biological systems in contact with them.

Antifouling coatings typically achieve antifouling effects through hydrophilic polymers (polyethylene glycol [PEG] and its derivatives [9], polysaccharides [10], polyamides [11], etc.) or zwitterionic polymers [12], which can prevent the adhesion of proteins and bacteria for a limited period. Bactericidal coatings generally function by inactivating bacteria attached to or suspended near the surface. They can be prepared by using silver nanoparticles [13] (AgNP), cationic polymers [14], antimicrobial peptides [15], and antibiotics [16] to functionalize the surface, or to modify the material or surface with quaternary ammonium salts (QAS) [17], graphene oxide (GO) [18], and show a certain bactericidal effect through release or contact killing. However, due to these limitations, a single bactericidal or antifouling material is hard to achieve ideal bactericidal effects. At present, there is no antifouling coating that can completely prevent bacterial adhesion. Such surfaces lack bactericidal function and cannot kill bacteria that adhere indiscriminately. Even if a small number of bacteria are attached, they will eventually grow into biofilms [19]. In addition, the bactericidal coating is confronted with the issue of its inability to efficiently eliminate dead bacterial cells and residues that quickly accumulate on the surface. This results in a deterioration of antibacterial effects, which even leads to the complete loss of bactericidal action. Conversely, these surfaces offer a platform for bacterial growth [20]. Therefore, developing a new coating with both antifouling and bactericidal properties has become an important research direction.

In recent years, diverse strategies to design surfaces with dual function of antifouling and bactericidal properties have been

employed, such as incorporating bactericides into antifouling surfaces via covalent immobilization, developing antifouling surfaces with controlled release of bactericides, and building a hierarchical structure that combines bactericidal and antifouling layers. Zwitterionic polymers show significant advantages in the field of antifouling and bactericidal properties due to their unique molecular structure and hydration capabilities. The zwitterionic polymer molecular chains have both positive and negative charge groups, which can form a highly hydrated super hydrophilic layer on the surface of material, which effectively resists the adsorption of proteins, bacteria, and other contaminants. Regarding bactericides, the most effective bactericidal agent usually contains cationic compounds, among which QAS polymers are the most widely studied. The QAS polymers not only demonstrate highly efficient killing ability against planktonic bacteria in solutions but also can effectively eliminate bacteria adhering to the material surface after being immobilized on the surface through specific methods, significantly reducing the adhesion, and proliferation of bacteria on the material surface. Previously, our group successfully constructed an antifouling and bactericidal surface containing zwitterionic ions and QASs, which showed excellent bactericidal properties [21].

However, directly constructing a stable zwitterionic coating on the material surface remains challenging, mainly due to issues like weak interface bonding between the coating and the substrate, complex preparation process, and poor thickness control [22]. Dopamine (DA)-assisted deposition technology is a surface modification method inspired by mussel adhesion proteins, which has attracted widespread attention in the field of materials science. DA molecules have a unique chemical structure that can spontaneously oxidize and polymerize under mild conditions, forming a strong polydopamine (PDA) coating on the surfaces of various materials. This coating not only displays fine adhesion but also provides an ideal platform for subsequent functional modification. The introduction of other functional molecules through chemical or physical action [23] can not only achieve rapid construction of the coating, but also further enhance the stability and functionality of the coating. For example, Shi et al. [24] modified the poly(vinylidene fluoride) (PVDF) bilayer hollow fiber membrane by coating DA and grafting polyethylene mine and then obtained a hydrophilic bactericidal PVDF membrane through quaternization reaction. The resulting membrane possessed ideal stability and long-term antibacterial property. Liu et al. [25] used PDA coating to prepare a polyacrylic brush anti-fouling reverse osmosis (RO) membrane by atom transfer radical polymerization. A PDA coating was first deposited on the RO membrane, and then the initiator BIBB was introduced into the DA coating. The film showed good water permeability and antifouling properties after modification. Based on this characteristic, the co-deposition of zwitterionic polymers assisted by DA as a mediator is thought to be a highly efficient and intelligent surface modification strategy.

Herein, we explored the introduction of DA-assisted deposition of zwitterionic polymers to enhance the adhesion of the coating. In this work, G20D80-5DMC was synthesized by free radical polymerization. After hydroxylation pretreatment, stainless steel (SS) as the substrate was immersed in a mixed solution of G20D80-5DMC and DA. Copper sulfate/hydrogen peroxide ($\text{CuSO}_4/\text{H}_2\text{O}_2$) as a catalyst was introduced to investigate its

effect on the oxidative polymerization rate of DA. The optimal co-deposition time was determined and the surface morphology, chemical composition, and hydrophilicity of the coating were characterized. Furthermore, the adhesion strength between the coating and the substrate was evaluated, and the coating's inhibitory effect and the bactericidal rates were quantitatively analyzed. Furthermore, the synergistic mechanism between [2-(methacryloyloxy) ethyl] dimethyl-(3-sulfopropyl) ammonium hydroxide (DMAPS) (zwitterionic groups) and [2-(methacryloyloxy) ethyl] trimethylammonium chloride solution (DMC) (quaternary ammonium cationic groups) was proposed. This not only provides a technical scheme for the rapid preparation of high-performance coatings with DA assistance, but also offers theoretical basis and technical support for biofouling prevention and control in fields such as marine engineering and biomedicine.

2 | Materials and Methods

2.1 | Materials

DMAPS was purchased from Shanghai Xinggui Biotechnology Co. Ltd., China. Glycidyl methacrylate (GMA) was purchased from Tianjin Heowns Biochem Technologie Co. Ltd., China. Triethylamine (TEA), DMC, DA, bovine serum albumin (BSA), $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$, and H_2O_2 were purchased from Shanghai Aladdin Reagent Co. Ltd., China. 2,2'-Azobis(2-methylpropionitrile) (AIBN) and 3-mercaptopropionic acid (3-MPA) were purchased from Shanghai Maclin Biochemical Technology Co. Ltd., China. Tris(hydroxymethyl) amino-methane hydrochloride was purchased from Tianjin Kemiu Chemical Reagent Co. Ltd., China. *Staphylococcus aureus* (*S. aureus*) and *Escherichia coli* (*E. coli*) were purchased from Beijing Microbiological Culture Collection Center. Nutrient Broth (NB) and Nutrient Agar Medium were purchased from Qingdao Hope Bio-Technology Co. Ltd., China. Methanol (MeOH) was purchased from Tianjin Fuyu Chemical Co. Ltd.,

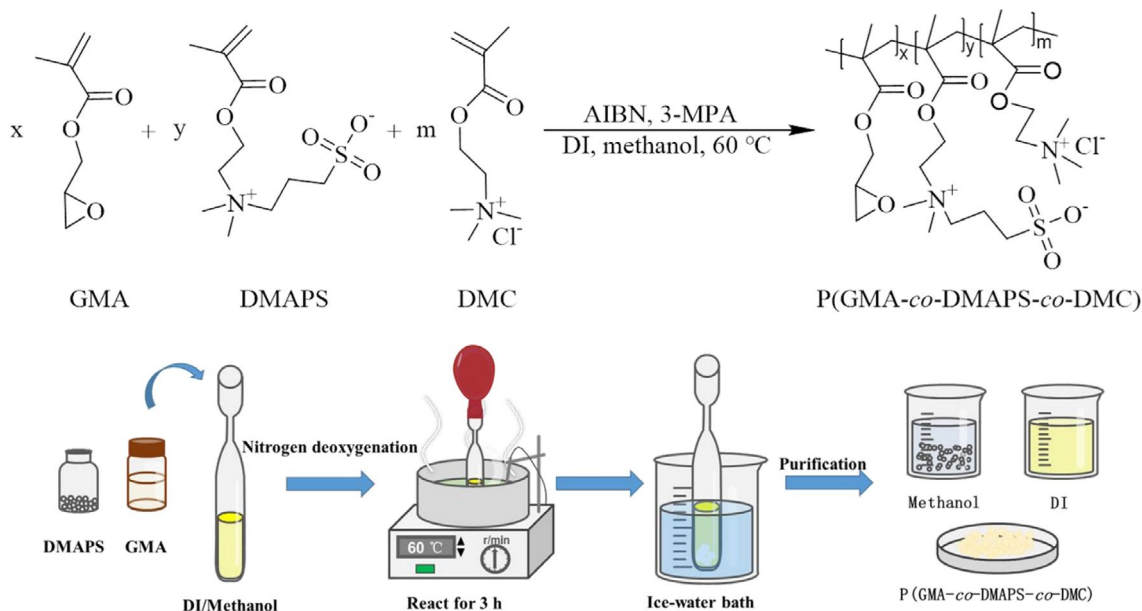
China. SS was purchased from Xinghua Yongsheng Stainless Steel Metal Products Co. Ltd., China.

2.2 | Synthesis and Characterizations of Polymer

The synthesis of P(GMA-co-DMAPS-co-DMC) was reported previously [21] (Scheme 1), and the optimal conditions were selected. A typical procedure is described as follows: First, GMA (20 mmol) was dissolved in 50 mL of methanol, as well as DMAPS (80 mmol) and DMC (3 mmol) in 50 mL of water. Then the two solutions were mixed and stirred for 15 min, and then purged with nitrogen for 30 min. The initiator AIBN (1 mmol) and the chain transfer agent 3-MPA (5 mmol) were added and stirred appropriately to dissolve thoroughly. The resulting solution was placed in an oil bath at 60 °C and stirring for 3 h. After the reaction was completed, the solution was placed in a cold-water bath for more than 3 h. The precipitate was purified with methanol three times, and dried to obtain a white crystalline powder, denoted as G20D80-5DMC. The control group polymer P(GMA-co-DMAPS) was prepared at the same feed ratio and was denoted as G20D80. The composition and chemical structure were characterized by ^1H NMR and Fourier transform infrared spectroscopy (FTIR), and the results were consistent with the previous report [21].

2.3 | Surface Modification and Characterization

The surfaces of SS were hydroxylated as described previously [21]. The surface of SS was polished by 240#, 400#, 800#, 1200#, or 2000# sandpapers, then washed with water and ethanol for 15 min, and finally rinsed with water and dried for later use. The as-synthesized G20D80-5DMC (50 mg/mL) and DA (2 mg/mL) were dissolved in Tris buffer solution (pH = 8.5), and $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ (5 mmol/L) and H_2O_2 (19.6 mmol/L) were added as oxidants. The substrates sanded with different grits were soaked in a mixed solution, and a layer of epoxy zwitterionic polymer



SCHEME 1 | Synthetic route of P(GMA-co-DMAPS-co-DMC) [21]. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/app.58083)]

coating was deposited on the surface, denoted as SS-PDA/PG20D80-5DMC.

The surface morphology and composition of the coatings were determined by scanning electron microscopy and X-ray photoelectron spectroscopy (XPS, Shimadzu AXISSUPPA, Japan), while the static water contact angle (WCA) of the coating surfaces was measured using an optical contact angle measuring instrument (Shanghai Xuanzhun, SZ-CAMC33, China).

2.4 | Adhesion Properties

The coating contains catechol functional groups, which imparts excellent bonding properties. Therefore, the bonding force on the SS surface was characterized by a universal tensile testing machine. The specimens used for overlap shear tests were prepared following the procedure: First, the SS sheets were cleaned with water and ethanol in sequence, and a certain amount of polymer solution was coated to the surface (20 mm × 20 mm). The two SS sheets were arranged in the same direction and the solution coated area was overlapped. The specimen was pressed and fixed with a pressure of 10 N for 12 h, and the specimen was stretched to a tensile speed of 5 mm/min until separation. The test was repeated on three parallel samples in each group. The shear strength (τ) is calculated from the maximum shear force in the stress-strain curve.

2.5 | Protein Adhesion

The anti-protein adhesion was determined by UV-visible spectroscopy. The protein solution displays a UV absorption peak at 275–280 nm. Within a certain concentration range, the absorbance of the protein solution is proportional to its concentration at the maximum absorption wavelength. BSA was used as a typical protein used in the experiment. The absorbance of solutions with different concentrations at a wavelength of 280 nm was measured by UV-visible spectrophotometer to draw a protein standard curve.

The samples were washed with phosphate-buffered saline (PBS, pH = 7.4) and immersed for 6 h. After that, the samples were put into the BSA solution and cultured at 25°C for 24 h. After incubation, the samples were gently rinsed with DI water and placed in a beaker for 20 min by sonication to fully desorb the BSA. The absorbance of desorption solution was measured using a UV-visible spectrophotometer at a wavelength of 280 nm for calculating the adsorption amount of BSA. The result was the average value of three parallel samples.

2.6 | Bactericidal Test

S. aureus and *E. coli* were selected to characterize anti-bacterial adhesion effect of the polymer coating. The semi-oscillating flask method was used to quantitatively analyze the bactericidal properties. The samples were sterilized by soaking in 75 wt. % alcohol and washed with PBS solution to remove residual alcohol and dried. Experimental samples were immersed in 5 mL of bacterial solution of 10^5 – 10^6 CFU/mL and oscillated

for 24 h at $36^\circ\text{C} \pm 1^\circ\text{C}$. 0.1 mL cultured bacterial solution was taken and inoculated on a plate, then viable bacteria were counted. Meanwhile, a blank sample was taken as the positive control group, and the identical batch of 0.9 wt. % NaCl solution and culture medium was taken as the negative control group. Experiments were repeated three times for each sample to calculate the average bactericidal rate.

3 | Results and Discussion

3.1 | Coating Preparation

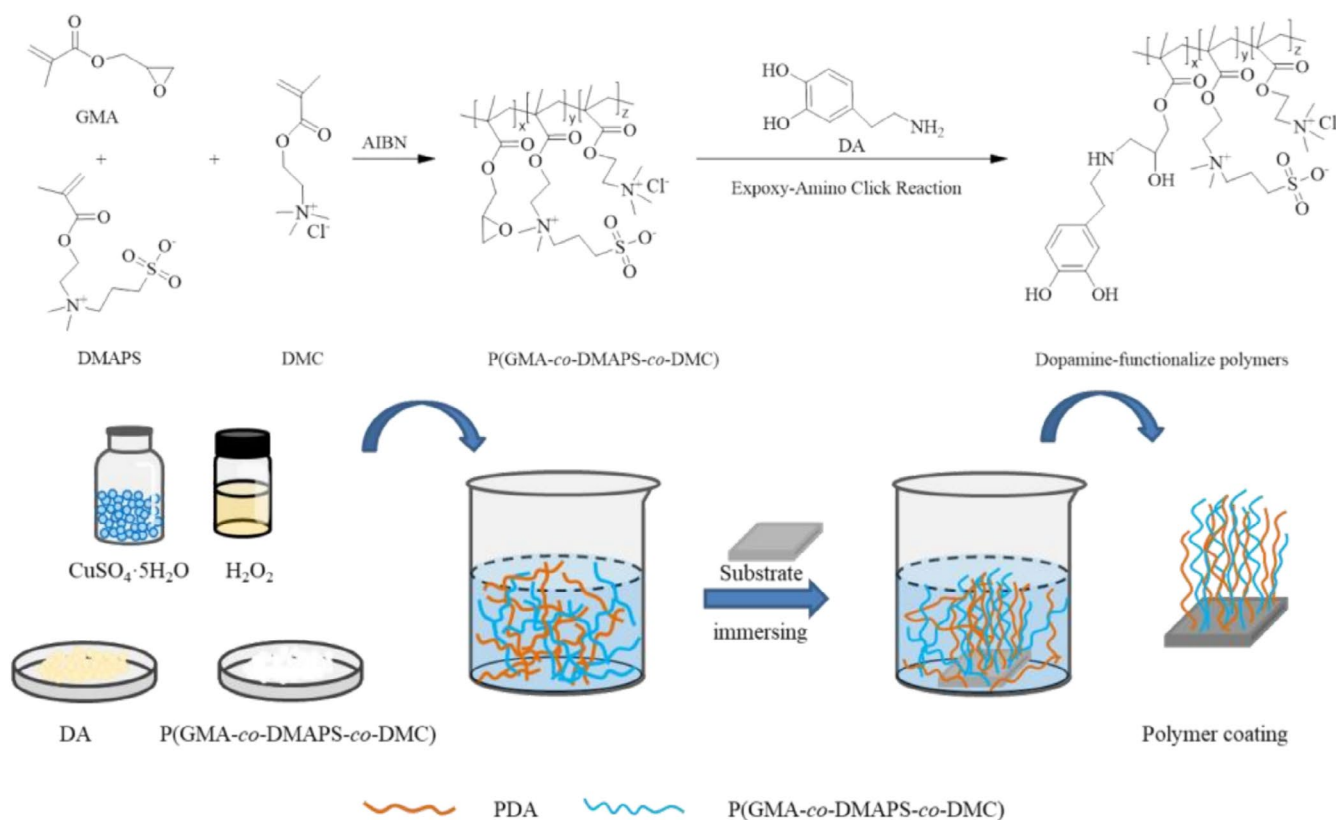
The coating achieves the co-deposition of polymers by leveraging the unique properties of DA. On the one hand, DA is converted into a quinone structure under an alkaline environment. Then, 5,6-dihydroxyindole is generated through cyclization and oxidation rearrangement, which eventually undergoes self-polymerization to form PDA. On the other hand, it can act as an anchor for deposition, and covalently bond with the polymer containing epoxy to assist the polymer to adhere to the material surface. G20D80 or G20D80-5DMC can achieve co-deposition with DA through epoxy-amino click reaction (Scheme 2).

3.1.1 | Catalyst Effect on Polymerization

DA was added to the Tris solution and exposed to air for a period. The color of the solution changed from colorless to light orange, indicating that DA is easily oxidized under alkaline environments. Since the oxidation of DA plays a crucial role in the formation of PDA coatings, the effect of oxidants on this process has been further studied. An experiment was conducted, in which one group of samples with $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}/\text{H}_2\text{O}_2$ as catalysts, while the other group without catalyst for comparison (Figure 1). As can be seen from the results, the solution catalyzed with oxidant turned black after 30 min, but without catalyst it only turned a light orange color, which shows that $\text{CuSO}_4/\text{H}_2\text{O}_2$ can promote the oxidative polymerization of DA, which is essential for achieving the desired properties in the coating preparation.

The oxidative polymerization of DA is quite complex and is commonly considered to be a free radical process. The $\text{CuSO}_4/\text{H}_2\text{O}_2$ system can produce substantial reactive oxygen free radicals, which can induce DA oxidation; on the other hand, the redox potential at which DA is transformed into quinone stands at 0.12 V. In comparison, Cu^{2+} with a higher redox potential can oxidize DA (0.34 V relative to the standard hydrogen potential). Therefore, it is essential to conduct a detailed investigation into the influence of $\text{CuSO}_4/\text{H}_2\text{O}_2$ on the oxidative polymerization behavior of DA, which can be beneficial to control the experimental process. Taking the deposition of DA/G20D80-5DMC as an example, the color changes in air, H_2O_2 , CuSO_4 , and $\text{CuSO}_4/\text{H}_2\text{O}_2$ were studied respectively, and the degree of oxidative polymerization was analyzed with the aid of characteristic peaks in the ultraviolet/visible spectra.

As seen from Figure 2a, in air, the color of the DA solution only changed from colorless transparent to light orange, dark brown,



SCHEME 2 | Mechanism of dopamine-assisted deposition. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/app.58083)]

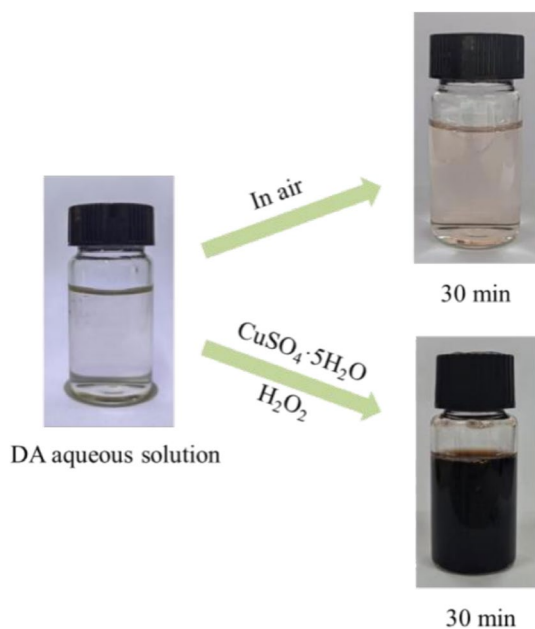


FIGURE 1 | Color changes of DA aqueous solution after 30 min with or without catalyst. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/app.58083)]

and black after 1, 8, and 18 h, respectively. With the addition of H_2O_2 , the solution turned light pink and light orange after 1 and 8 h, respectively, and gradually deepen with lasting time. The slow color changes indicate that the introduction of H_2O_2 inhibits the oxidative auto-polymerization of DA to a certain extent

[26]. With the addition of CuSO_4 , the solution quickly turned green and brown, proving that Cu^{2+} can act as an oxidizing agent. With the addition of CuSO_4 and H_2O_2 at the same time, the solution turned dark red in a short time, and the color of the solution gradually deepened until it was dark black, indicating that DA is rapidly oxidized and polymerized to form PDA. This significant and rapid color change indicated that DA underwent rapid oxidation and polymerization to form PDA. By comparing the color change rates and the final states under different conditions, it can be clearly demonstrated that the combination of CuSO_4 and H_2O_2 can greatly accelerate the oxidative polymerization process of DA.

During the oxidation process of DA, a large amount of indole structures generated, resulting in an obvious characteristic peak at 420 nm in the UV-visible spectrum. To further analyze the effect of the catalyst on the reaction, the oxidation status of the reaction solution will be evaluated based on the intensity of the characteristic peaks of the intermediate products. Therefore, the absorbance at 420 nm of DA/G20D80-5DMC aqueous solution at different reaction time in presence of air, H_2O_2 , CuSO_4 , and $\text{CuSO}_4/\text{H}_2\text{O}_2$ were compared respectively to further analyze the effect of the catalyst. As shown in Figure 2b, the oxidation rate of DA was relatively slow in air, and the solution color turned black with an absorbance of 1.24 after 24 h. With the addition of H_2O_2 , the absorbance was 0.73, smaller than that in the air after 24 h. With the addition of CuSO_4 , the solution turned black with an absorbance of 0.81 at 8 h, which further demonstrates that CuSO_4 plays an important role in promoting the oxidation and polymerization of DA. Overall, these results indicate that CuSO_4 and H_2O_2 have distinct impacts on

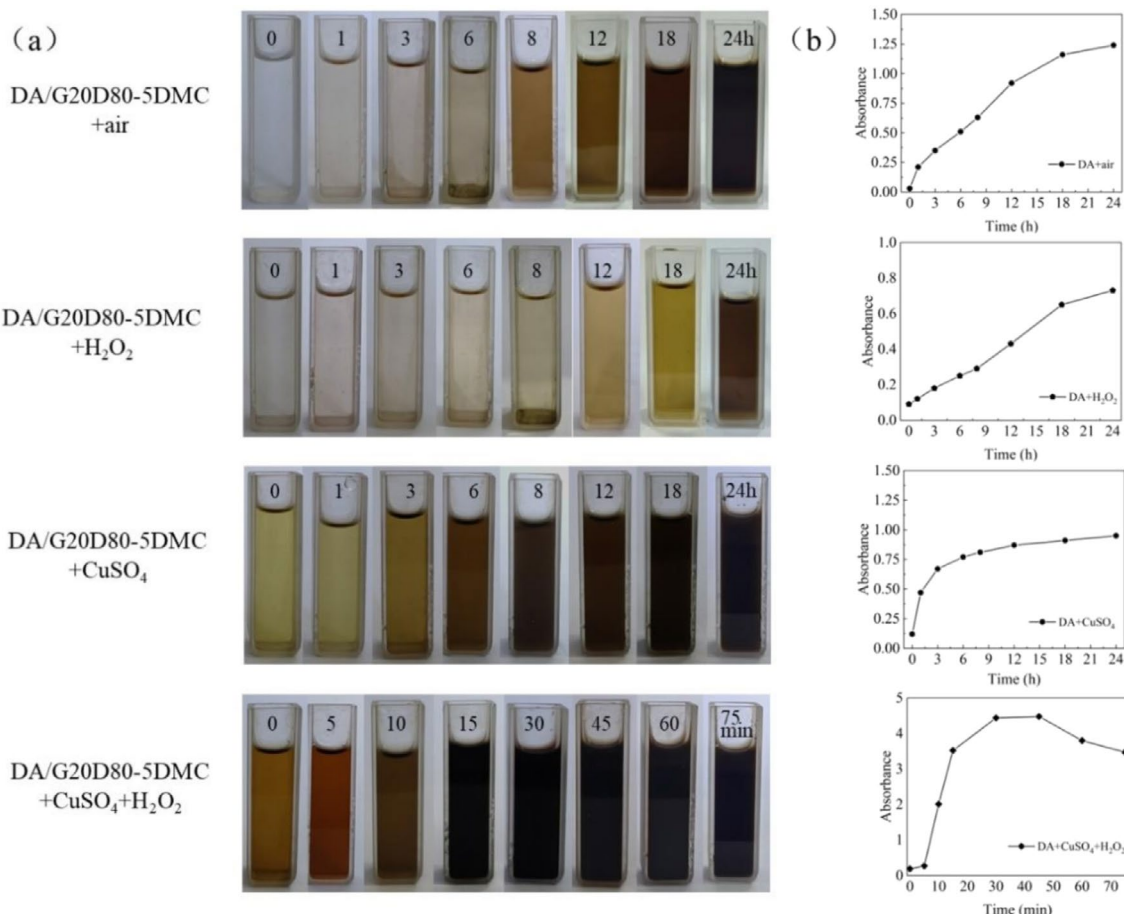


FIGURE 2 | (a) Diagrams at different times; (b) Absorbance at 420 nm over reaction solution under different conditions). [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/app.58083)]

the oxidative polymerization of DA, and further investigation is needed to fully understand the underlying mechanisms.

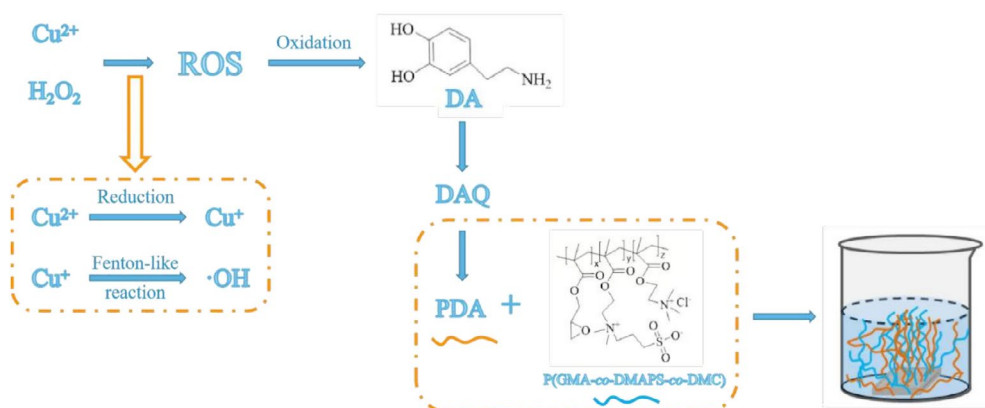
By comparison of the absorbance at the same time, the absorbance of DA/G20D80-5DMC aqueous solution under air, H₂O₂, CuSO₄, and CuSO₄/H₂O₂ were 0.21, 0.12, 0.47, and 3.8 after 1 h, respectively, which are consistent with the color change of the solution.

The results indicate that CuSO₄/H₂O₂ plays a synergistic role in accelerating the oxidation behavior of DA/G20D80-5DMC, because CuSO₄ and H₂O₂ can react in a way like the Fenton reaction with generating many reactive oxygen free radicals, which are highly reactive and can readily attack DA molecules, thereby accelerating the oxidation and polymerization of DA. In this system, Cu²⁺ exhibits a certain effect on DA polymerization. However, when CuSO₄ is used alone as an oxidizing agent, the oxidation rate is relatively slow. Upon the addition of H₂O₂, Cu²⁺ in CuSO₄ reacts with H₂O₂ to generate Cu⁺. Under the experimental conditions where H₂O₂ is in excess, Cu⁺ in the solution further reacts with H₂O₂ in the reaction system, producing Cu²⁺ and ·OH free radicals. A large number of ·OH free radicals in the solution then oxidize DA, triggering its self-polymerization. During the co-deposition process, H₂O₂ serves as the main source of free radicals, while CuSO₄ activates and

enhances this process through valence cycling and chelation. Both substances are indispensable for achieving polymer deposition. The proposed catalytic pathway involving CuSO₄/H₂O₂ was shown in Scheme 3.

3.1.2 | Dopamine Polymerization Behavior

The oxidative polymerization state of DA in solutions plays a critical role in the formation of DA co-deposition coating (Figure 3). At 6 h, many precipitates were produced in the DA solution with oxidizing agents. This is because the PDA in the solution gradually assembled into large aggregates and precipitated. With the addition of G20D80/G20D80-5DMC, the solution turned black without precipitation, indicating that the oxidative polymerization of DA still took place. However, the absence of precipitation implies that the normal process of aggregation and assembly of PDA was disrupted. These phenomena indicate that the introduction of G20D80/G20D80-5DMC affected the oxidative auto-polymerization of DA and weakened the aggregation and assembly of polymers. The polymer with DMAPS units can result in an increase of hydrophilicity and destroy the interaction force between PDA aggregates. This demonstrates that an appropriate amount of G20D80/G20D80-5DMC can effectively reduce the aggregation and settlement of PDA aggregates and



SCHEME 3 | The proposed catalytic pathway involving $\text{CuSO}_4/\text{H}_2\text{O}_2$. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/app.58083)]

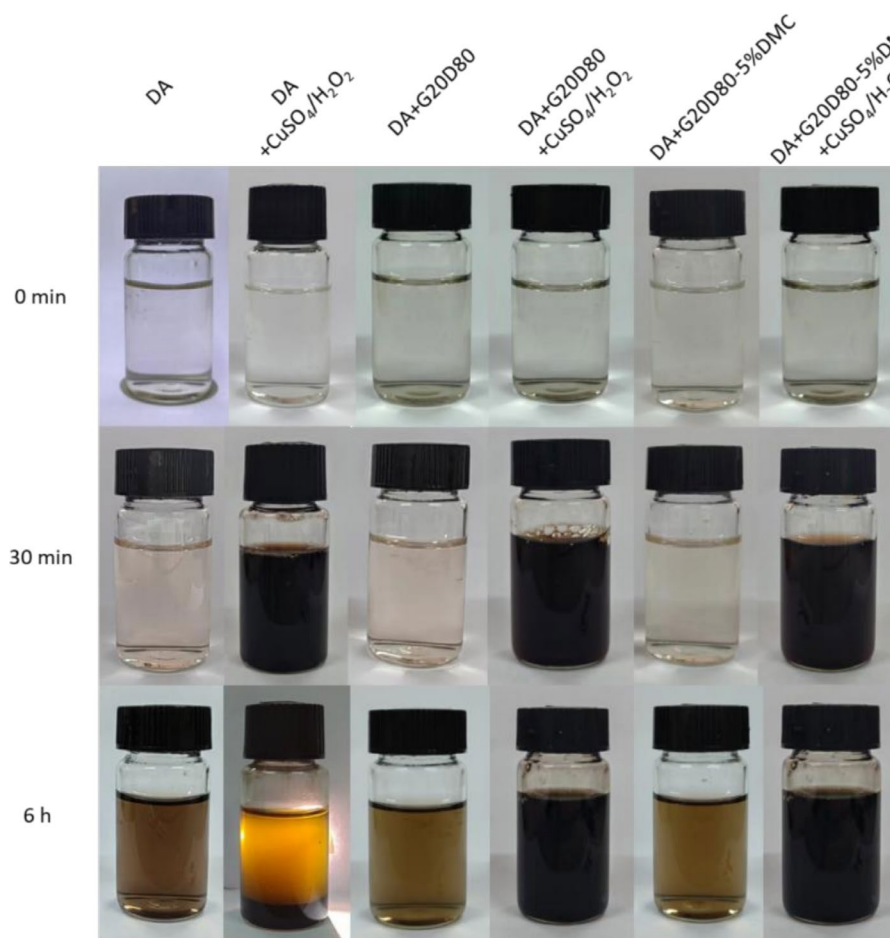


FIGURE 3 | Color changes of DA, DA/G20D80, and DA/G20D80-5DMC mixed solutions with or without oxidant. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/app.58083)]

increase the utilization efficiency of aggregates in the solution. More aggregates can deposit on the surface of substrate instead of precipitating in the solution.

3.1.3 | Modification Time

The appropriate grafting time can make the resulting surface more uniform and flatter. With the $\text{CuSO}_4/\text{H}_2\text{O}_2$ catalysis,

the degree of polymerization can be estimated by the absorption peak of the DA oxidation intermediate indole in the UV/vis spectrum to determine the optimal time for G20D80 and G20D80-5DMC to form the coating in DA solution, which is specifically shown in Figure 4. The absorbance in the DA solution rose rapidly from 0.19 to 4.44 within 30 min, and after reaching a peak of 4.48 between 30 and 45 min, it showed a significant downward trend. After the addition of G20D80 or G20D80-5DMC, the time when the absorbance reached the

peak of ~ 3.5 slightly shifted backward, basically staying at about 45 min, and the absorbance was relatively stable after reaching the peak, without a significant downward trend. It is determined that the solution contains a large amount of PDI intermediates at 45 min, which is conducive to the deposition of DA and polymers.

3.2 | Surface Characterization

The surface morphology of SS-PDA and SS-PDA/G20D80-5DMC was shown in Figure 5. It can be observed that particles with different sizes formed on the surface, exhibiting obvious agglomeration (Figure 5a). DA is prone to over assembly during self-polymerization, resulting in particle agglomeration. When DA was jointly deposited with G20D80-5DMC, the surface was relatively flat with many nanoparticles (Figure 5b). The size and distribution of these nanoparticles are relatively uniform, with a particle size of about 150 nm. Compared with DA, the smaller particles display obvious independence. The introduction of G20D80-5DMC can affect the oxidative self-polymerization of DA by weakening the aggregation and assembly of products and destroying the interaction force between PDA aggregates. Meanwhile, the introduction of G20D80-5DMC plays a crucial role in altering

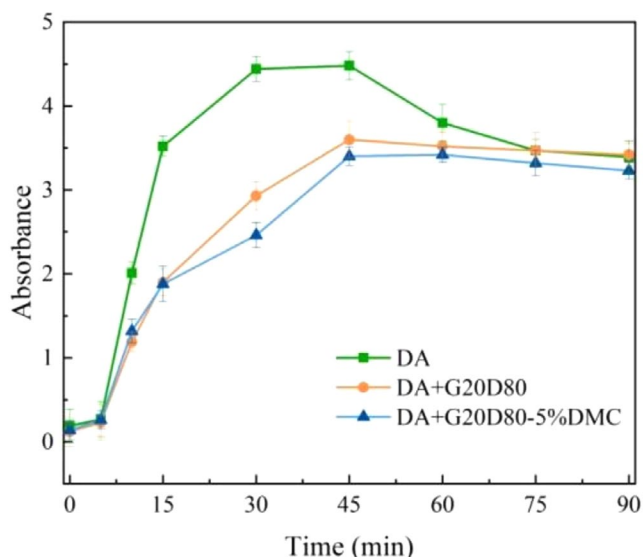


FIGURE 4 | Absorbance in different deposits with reaction time. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

the co-deposition. It weakens the aggregation and assembly of the products and disrupts the interaction between PDA aggregates. As a result, it not only enables PDA to be more fully utilized but also ensures its uniform deposition on the surface instead of rapid aggregation and deposition.

Figure 6 shows the full XPS spectra and C1s, N1s, S2p spectra of SS, SS-PDA, and SS-PDA/G20D80-5DMC, respectively. Several obvious peaks can be seen in the full XPS spectra, the photoelectric peaks at 284.8, 402.7, and 532.0 eV correspond to C1s, N1s, and O1s, respectively (Table 1). There is a new peak in SS-PDA/G20D80-5DMC at 168.2 eV, which can be attributed to the sulfonic acid group ($-\text{SO}_3\text{H}$) in the DMAPS fragment. The content of N element in SS-PDA and SS-PDA/G20D80-5DMC is significantly higher than that of the SS. The

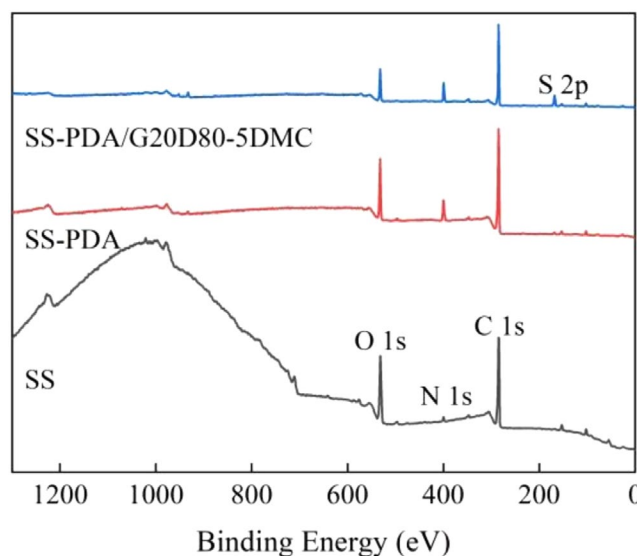


FIGURE 6 | XPS spectra of stainless steel, SS-PDA, and S-PDA/G20D80-5DMC. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

TABLE 1 | Element analysis of XPS.

Element (at.%)	C 1s	O 1s	N 1s	S 2p
SS	74.54	24.10	1.37	—
SS-PDA	68.29	24.28	7.44	/
SS-PDA/G20D80-5DMC	70.22	23.8	4.58	1.41

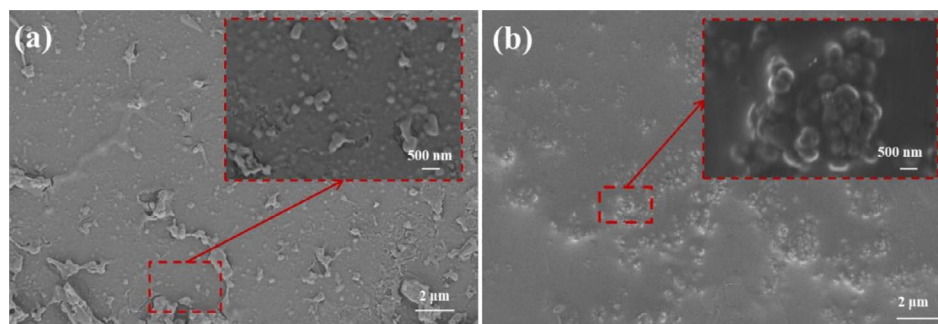


FIGURE 5 | Surface topography of SS-PDA (a) and SS-PDA/G20D80-5DMC (b). [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

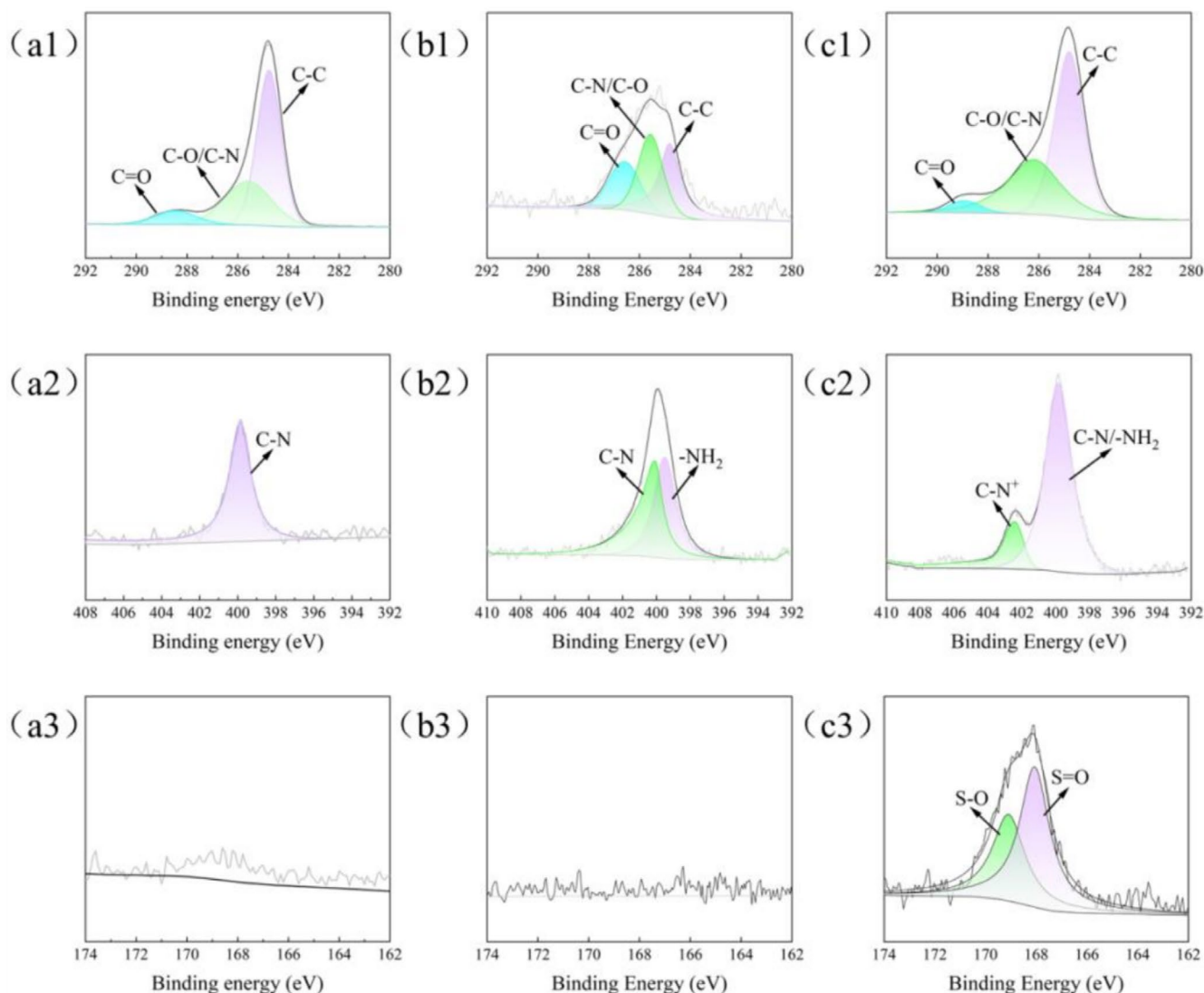


FIGURE 7 | Curve fitting of the XPS narrow scan spectra of stainless steel (a1–3), SS-PDA (b1–3), and S-PDA/G20D80-5DMC (c1–3). [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/app.58083)]

enhancement and appearance of these characteristic peaks are attributed to DA and zwitterionic polymers deposited on the surface.

To further understand the bonding types of functional groups of modified SS, the fine spectrum was fitted and analyzed, shown in Figure 7. Figure 7b shows the C1s and N1s fitting diagrams of SS-PDA. Three peaks at 284.8, 285.6, and 286.6 eV in the C1s spectrum of SS-PDA correspond to C–C, C–O/C–N, and C=O bonds, respectively. There are two peaks at 399.49 and 400.08 eV in N1s spectrum, corresponding to –NH₂ and C–N, respectively. Compared with SS, the existence of these peaks proves that the surface has been successfully modified by PDA. Figure 7c shows the C1s and N1s fitting diagrams of SS-PDA/G20D80-5DMC. In the C1s spectrum, the electron binding energy of C–C, C–O/C–N, and C=O bonds positions at 284.8, 286.18, and 288.9 eV, respectively. There are two peaks at 402.43 and 399.83 eV in N1s, corresponding to C–N⁺ and C–N/–NH₂. When performing the fitting here, these are fitted into one peak because the characteristic peak positions of C–N and –NH₂ are close. In addition, characteristic peaks

of sulfur appeared on this surface, and they were fitted into two peaks at 167.8 and 169.0 eV, corresponding to 2p_{3/2} and 2p_{1/2}, respectively. The ratio of peak area is 1.97:1, the ratio of peak and peak area to the characteristic fitting peak and peak area of sulfur element is 2:1, and the ratio of peak and half width and height is 1:1. In the fitting spectrum, the appearance of C–N⁺ and S demonstrates that zwitterionic polymers have been successfully deposited on the surface of SS with the assistance of DA.

3.3 | Adhesion Properties

As shown in Figure 8, the difference in adhesion between SS-G20D80, SS-PDA, SS-PDA/G20D80, and SS-PDA/G20D80-5DMC was compared. It was found that the adhesion strength of the coating doubled after the introduction of DA, and the surface adhesion strength of SS-G20D80 and SS-G20D80-5DMC was lower than 20 KPa. The adhesion strength is mainly since the epoxy groups in the GMA fragment bind to the hydroxyl groups on the substrate under alkali-induced epoxy ring-opening,

forming stable chemical bonds, and their adhesion strength is limited. With the addition of DA, the adhesion strength can reach 40 KPa by a large amount of catechol groups in DA, which can combine with the SS surface through hydrogen bonding and π - π stacking, forming bidentate chelation with the metal oxide on the SS. In addition, Cu^{2+} as catalyst can form a strong coordination bond with the catechol groups in PDA during the coating formation process, enhancing the interaction, and significantly improving adhesion.

3.4 | Antifouling and Bactericidal Performance

3.4.1 | Anti-Protein Adsorption Performance of Coatings

Figure 9a shows the WCAs with an order of SS-PDA > SS-PDA/G20D80 > SS-PDA/G20D80-5DMC. Compared the SS with a WCA of $62.4 \pm 1.6^\circ$, the WCA of SS-PDA surface drops to $37.1^\circ \pm 2.5^\circ$. A large amount of hydroxyl and amino groups in DA can bind with water molecules through hydrogen bonds,

resulting in a more hydrophilic surface. The SS-PDA/G20D80 surface and the SS-PDA/G20D80-5DMC surface are super hydrophilic with contact angles of $14.9 \pm 3.0^\circ$ and $11.8 \pm 2.1^\circ$, respectively, due to the introduction of zwitterionic monomers. The coating surface can bind with many water molecules through electrostatic hydration, which further enhances the hydrophilicity.

The relative protein adsorption rates of SS-PDA, SS-PDA/G20D80, and SS-PDA/G20D80-5DMC were 29.83%, 6.37%, and 6.61%, respectively (Figure 9b). The amount of protein adsorbed by SS-PDA/G20D80 and SS-PDA/G20D80-5DMC is significantly lower than that adsorbed by SS surfaces. Zwitterions with strong hydration capacity can block the contact between proteins and surfaces, thereby exhibiting good protein adhesion resistance. However, the poor protein adhesion of SS-PDA can be ascribed to catechol, amino, and hydroxyl functional groups, which may induce proteins to adsorb on the surface due to electrostatic interaction or hydrogen-bonding interaction. The deposition surface with uneven particle size and large roughness may lead to more easily adhesion of proteins.

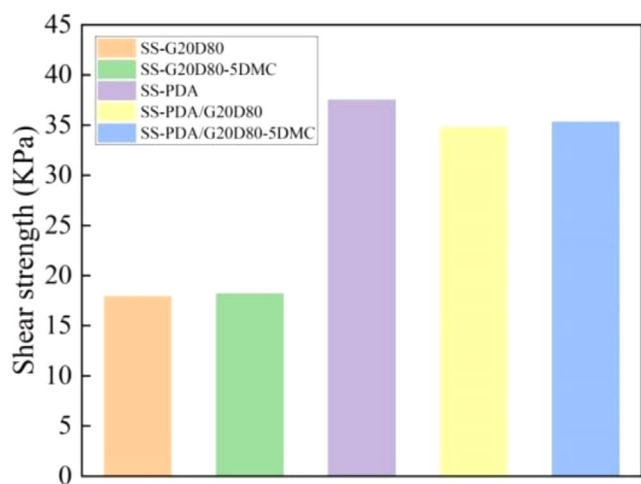


FIGURE 8 | Comparison of adhesion strengths of different surfaces. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

3.4.2 | Bacterial Attachment and Inactivation Performance of Coatings

To study the impact of DA on the bactericidal properties, the as-prepared coatings with or without DA were compared. As shown in Table 2 and Figure 10, SS-PDA and SS-PDA/G20D80 display low bactericidal ability for the two types of bacteria.

TABLE 2 | Number of colonies of *E. coli* and *S. aureus* on different surfaces.

	Blank	SS	SS-DA	SS-PDA/ G20D80	SS-PDA/ G20D80- 5DMC
<i>E. coli</i>	218	198	204	172	5
<i>S. aureus</i>	226	205	173	132	0

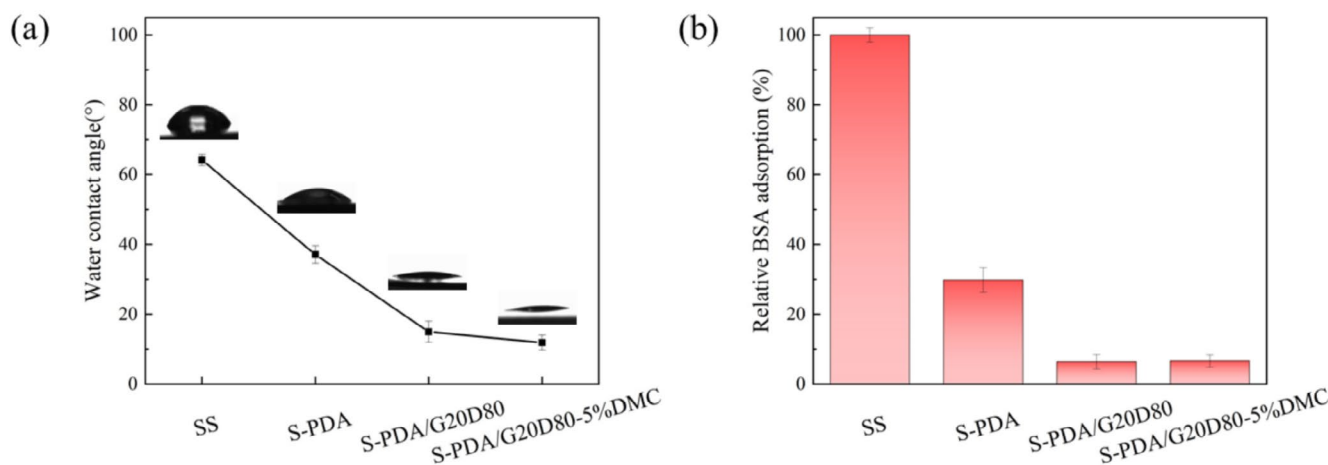


FIGURE 9 | Static water contact angles (a) and relative BSA adsorption (b) of different surfaces. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

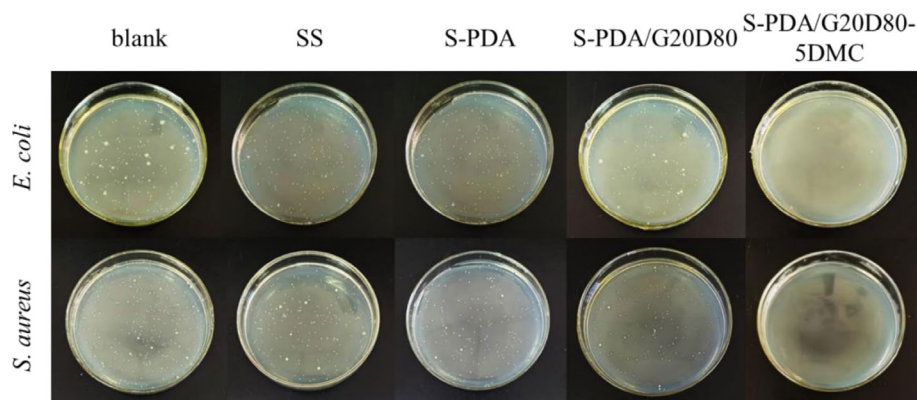


FIGURE 10 | Bactericidal properties of different surfaces to *E. coli* and *S. aureus*. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/app.58083)]

TABLE 3 | Number of colonies of *E. coli* and *S. aureus* on different surfaces.

	SS	SS-DA	SS-PDA/ G20D80	SS-PDA/ G20D80- 5DMC
<i>E. coli</i>	82	40	9	0
<i>S. aureus</i>	65	28	2	0

PDA and G20D80 without bactericidal properties cannot kill or inhibit bacteria through chemical action. The number of live bacteria in *E. coli* on the surface of SS-PDA/G20D80-5DMC decreased from 90.8% (number of live bacteria on the surface of original SS) to 2.3%, and the number of live bacteria in *S. aureus* decreased from 90.7% (number of live bacteria on the surface of original SS) to 0.01%. The positively charged quaternary ammonium groups in DMC can bind and destroy the cell wall, resulting in the death of bacteria. In contrast, the surface of SS-G20D80-5DMC exhibits excellent bactericidal property. In addition, CuSO_4 as a catalyst was added during the co-deposition process, and some of which will remain in the coating and display a partial bactericidal effect. For SS-PDA/G20D80-5DMC and SS-G20D80-5DMC, with little difference in bactericidal rate, co-deposition of G20D80-5DMC onto the surface assisted by DA is a feasible method with excellent anti-fouling and bactericidal effect.

The SS-PDA surface exhibited worse hydrophilicity than that containing zwitterionic polymers, which was evaluated through bacterial adhesion testing. As shown in Table 3 and Figure 11, the SS-PDA surface exhibits poor anti-bacterial adhesion (the adhesion rates of *E. coli* and *S. aureus* on the surface were 48.78% and 43.08%, respectively). Hydrophilic compositions on the SS-PDA surface like amino groups were consumed, leading to a reduction of the hydrophilicity during the reaction. As a result, an effective hydration layer failed to form and prevent bacterial adhesion. Moreover, the surface of SS-PDA may have a certain positive charge, which is prone to electrostatic attraction with the negatively charged bacterial cell membrane, thereby promoting bacterial adhesion. The amount of bacterial adhesion on the surface of SS-PDA/G20D80-5DMC was greatly reduced (The adhesion rates of *E. coli* and *S. aureus* on the surface were both

0) due to the formation of a hydrated layer by the combination of the coating and free water through electrostatic interaction, which can block the contact between bacteria and the surface, thereby can effectively resist bacterial adhesion.

3.5 | Synergistic Mechanism of Antifouling and Bactericidal

Based on the above-mentioned experimental results, it is proposed that SS-PDA/G20D80-5DMC exhibits effective antifouling and bactericidal synergy, which can be primarily attributed to the DMAPS and DMC functional groups. Proteins, as typical amphiphilic organic pollutants, maintain specific conformations in aqueous solutions through hydrogen bonding and electrostatic interactions with water molecules. The zwitterionic molecular chain comprises isoelectric units, each containing both positively charged groups and negatively charged groups. These units can bind up to eight water molecules via electrostatic interactions, forming a stable hydration layer that acts as a physical barrier to prevent direct contact between the surface and contaminants. This mechanism inhibits protein deposition by blocking their access to the coating surface. Furthermore, the electrically neutral character of the polymer reduces electrostatic interactions with the charged proteins, thereby minimizing excessive protein adsorption on the membrane surface.

The positively charged quaternary ammonium groups in the DMC segment can disrupt negatively charged microbial cell membranes through electrostatic attraction, leading to the leakage of cytoplasm and the death of bacteria. Previous studies indicate that the bactericidal efficacy of DMC correlates with its content and surface hydrophilicity. With abundant hydrophilic groups, the polymeric chains exhibit enhanced extension in aqueous environments, facilitating the contact between the bactericidal monomers and bacterial cells. Conversely, the reduced hydrophilicity diminishes both chain extension and bacterial adsorption/elimination rates. Notably, SS-PDA/G20D80-5DMC with numerous DMAPS groups possesses high hydrophilicity to promote rapid molecular chain extension in water. This structural feature not only enhances DMC segment mobility for improved bacterial contact but also suppresses biofilm formation by residual dead bacteria. The complementary functions of

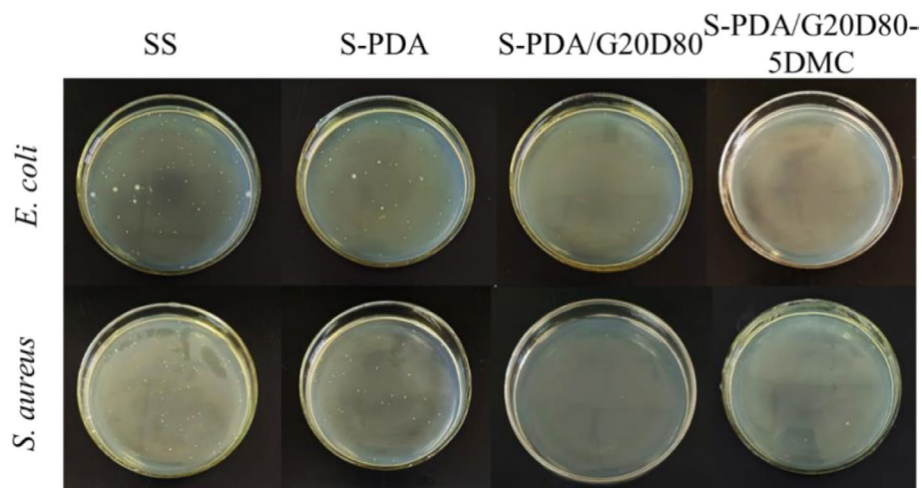
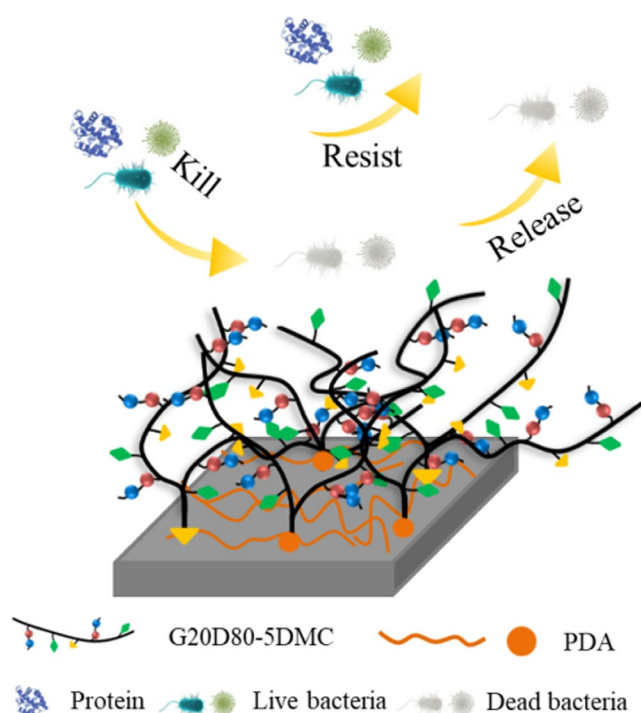


FIGURE 11 | Adsorption of *E. coli* and *S. aureus* on different surfaces. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/app.58083)]



SCHEME 4 | Synergistic mechanism of antifouling and bactericidal. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/app.58083)]

DMAPS and DMC collectively achieve superior antifouling and bactericidal synergy (Scheme 4).

4 | Conclusions

In this work, a simple, stable, and versatile method, that is, DA-assisted deposition of polymers to prepare antifouling and bactericidal coatings, has been designed. DA was introduced as the anchor for deposition, and G20D80-5DMC was deposited onto the substrate surface with the assistance of DA to improve the adhesion to the surface. The addition of catalysts induced rapid oxidative polymerization of DA and shortened the co-deposition time to 45 min. SS-PDA/G20D80-5DMC with super hydrophilicity displayed a surface static WCA of 5.3°, a protein adhesion

resistance rate of 93%, and bactericidal rates of 97.7% and 99.9% for *E. coli* and *S. aureus*, respectively. On the surface of coating, the zwitterionic groups and cationic groups played a synergistic role, exerting excellent antifouling, and bactericidal properties, which will provide a new idea and method for the subsequent development of high-performance antifouling and bactericidal coatings.

Author Contributions

Dejie Zhou: data curation (lead), formal analysis (lead), writing – original draft (lead). **Fei Ma:** data curation (equal), formal analysis (equal), investigation (equal), visualization (lead), writing – original draft (equal). **Xuanjie Zheng:** investigation (supporting). **Nianfeng Han:** formal analysis (supporting), investigation (supporting). **Xinde Tang:** conceptualization (equal), project administration (lead), writing – review and editing (lead). **Fuying Dong:** conceptualization (supporting), investigation (supporting). **Haixia Cheng:** conceptualization (supporting), formal analysis (supporting), resources (equal). **Peng Wang:** conceptualization (supporting), investigation (supporting). **Qingzhao Wang:** conceptualization (equal), methodology (equal), supervision (equal), writing – review and editing (supporting).

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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